Seminar: Large Language Models

# Textbooks Are All You Need

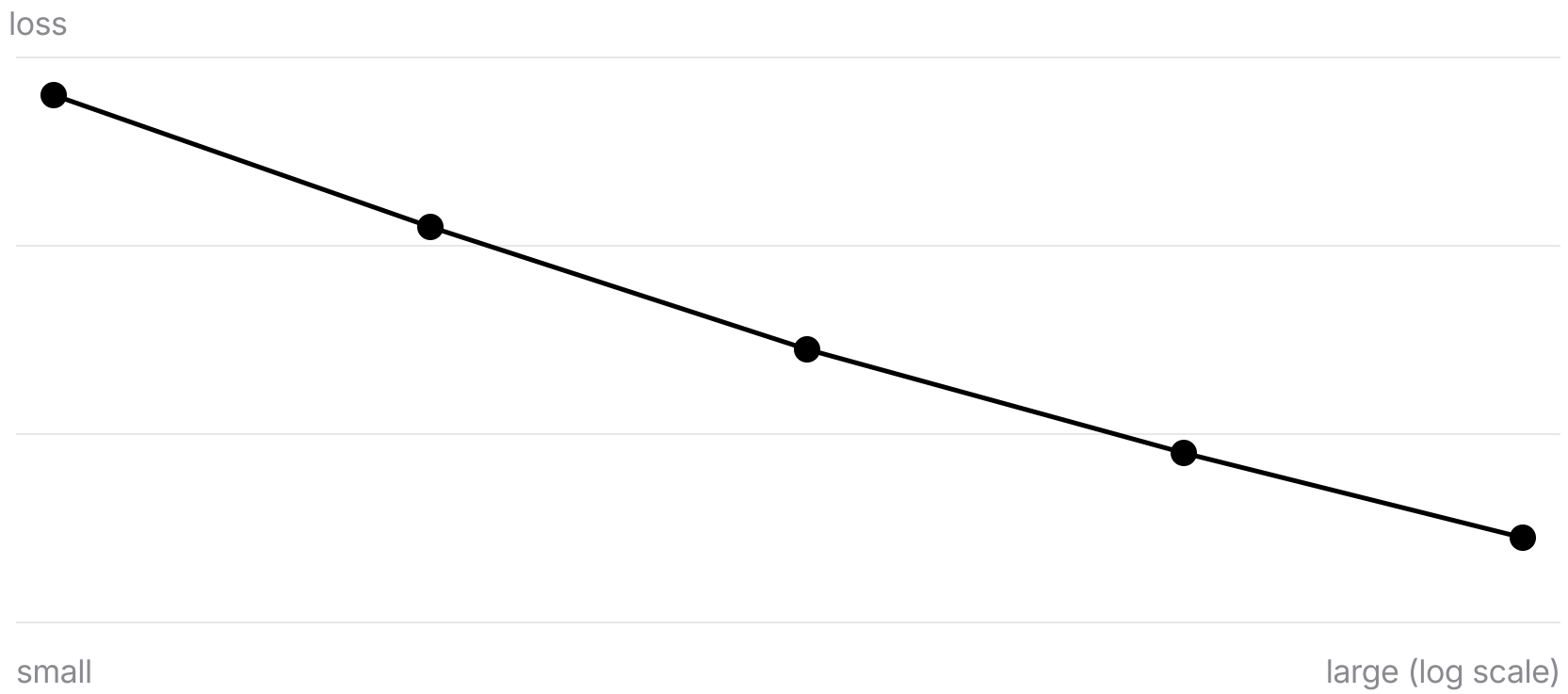
Gunasekar et al. · Microsoft Research · 2023

Bartosz Kruszewski

# What if better data beats bigger models?

# Scaling laws

more parameters, more compute → lower loss




Kaplan et al., 2020 · Hoffmann et al., 2022 (cited in the paper) · TinyStories: Eldan & Li, 2023

# The domain: code

write a Python function from its docstring

```python
def incr_list(l: list):
    """Return list with elements
    incremented by 1."""
    return [x + 1 for x in l]
```

**HumanEval**

164 problems · unit tests

**MBPP**

mostly basic Python problems

**pass@1**

one try, must pass all tests

# phi-1 at a glance

| 1.3B | 7B | 4 days | 50.6% |
| --- | --- | --- | --- |
| parameters | tokens of data | on 8 × A100 | HumanEval pass@1 |

# HumanEval vs open & closed models

pass@1, self-reported · June 2023

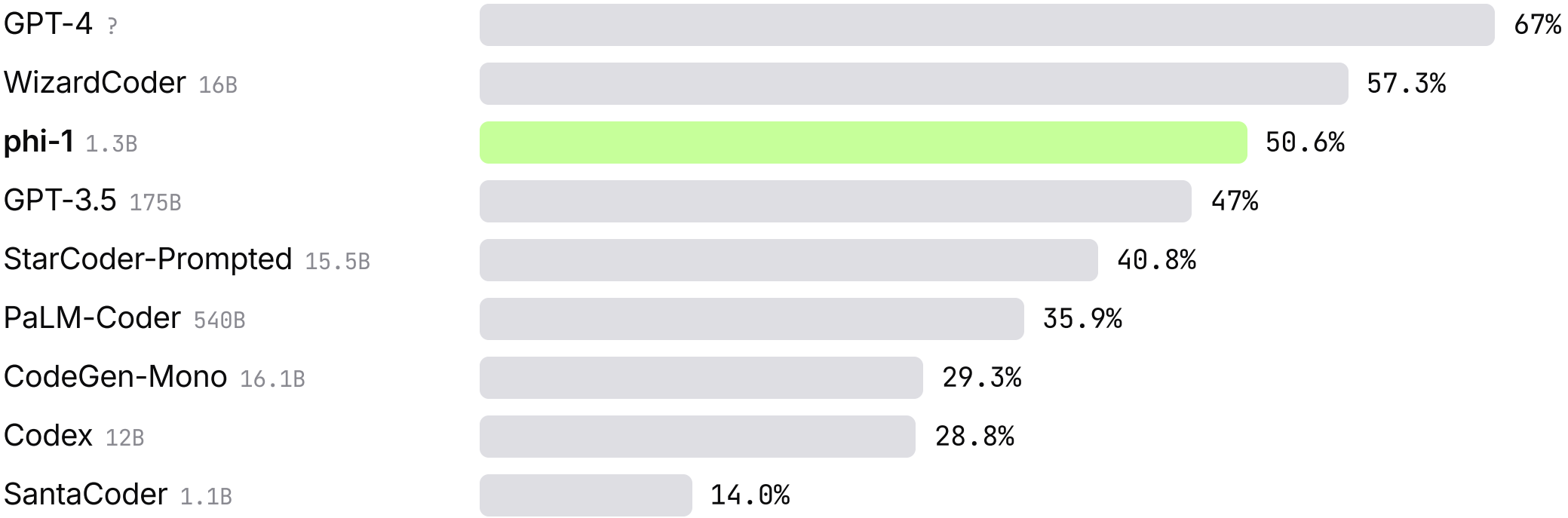



Table 1 of the paper

# Size vs score

HumanEval pass@1 vs parameters

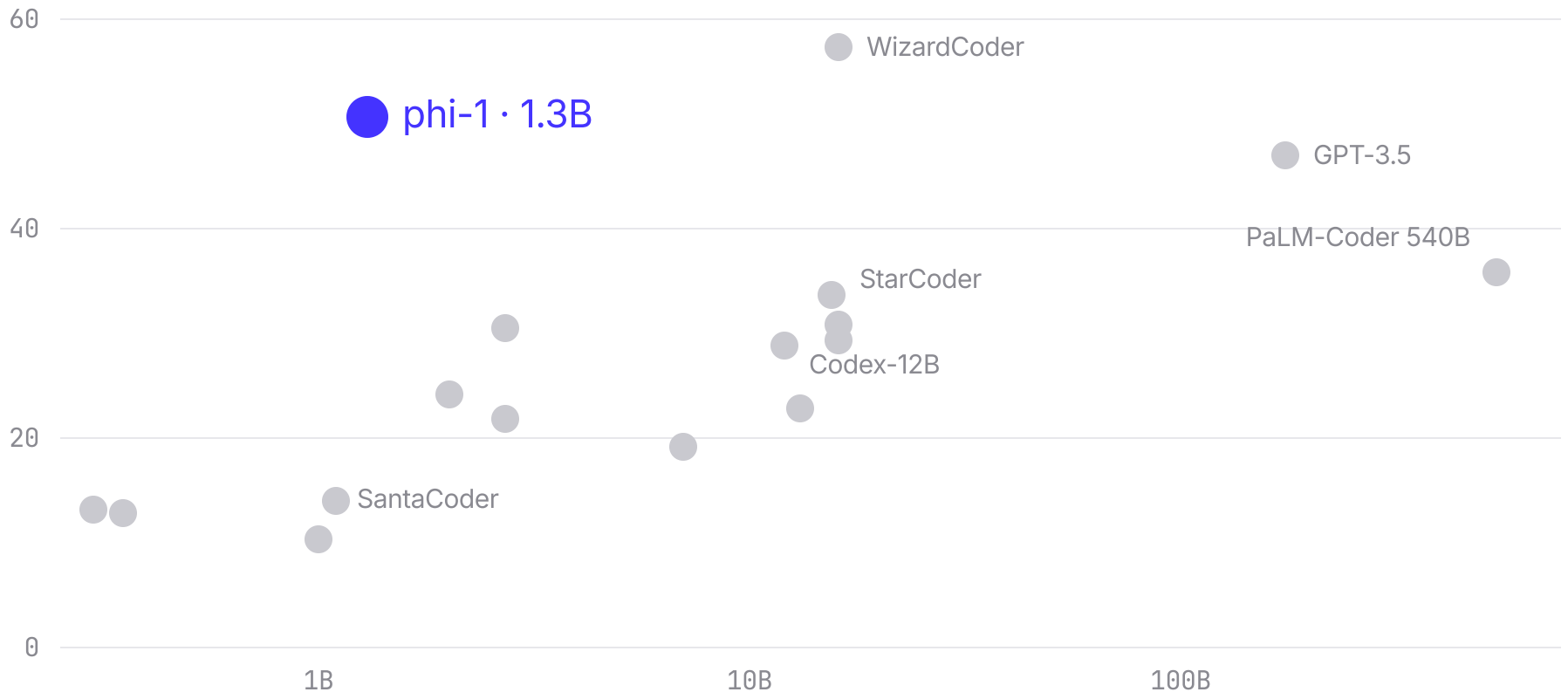



Table 1 · x axis log scale

# What's wrong with web code?

The Stack · StackOverflow — huge, diverse, but not instructive

01

## Not self-contained

depends on other files

02

## Boilerplate

constants, configs, GUI setup

03

## Buried logic

inside long, undocumented functions

04

## Skewed topics

unbalanced coverage of skills

# High vs low educational value

✓ high educational value

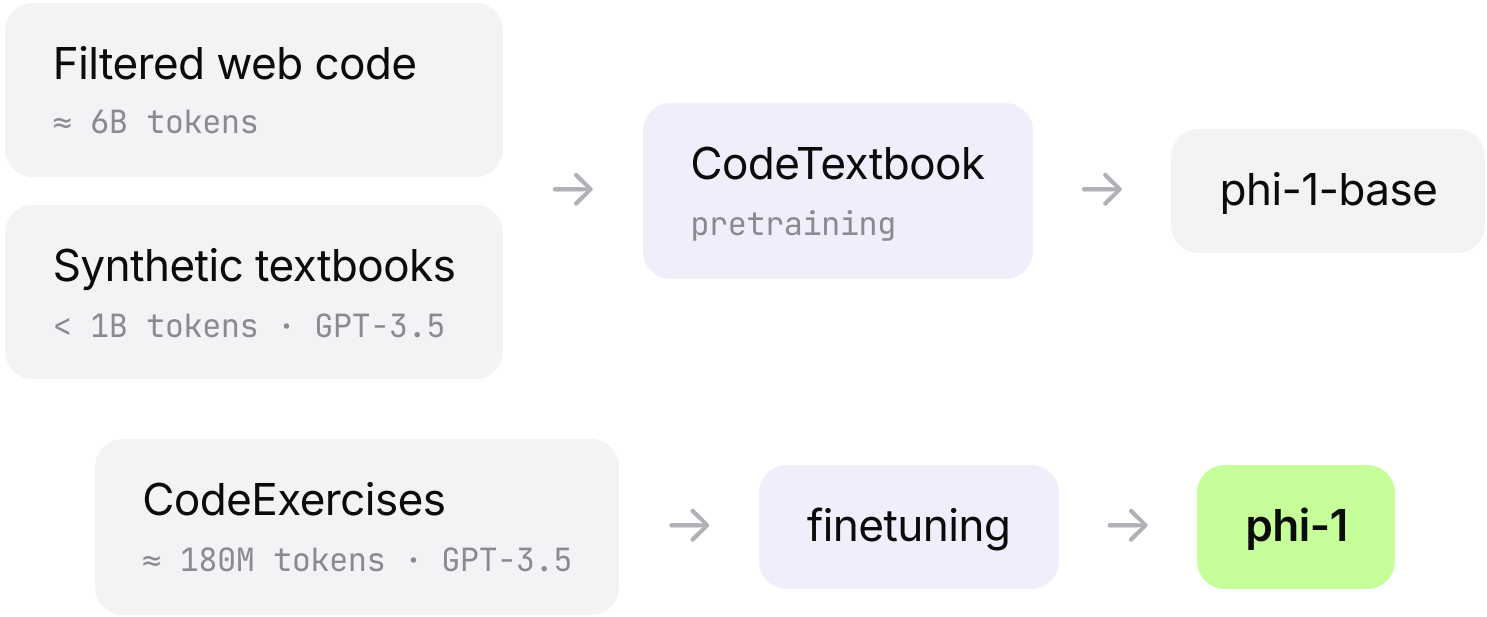
```
def cosine_dist(x, y):
    """Computes Cosine Distance."""
    x = F.normalize(x, dim=1)
    y = F.normalize(y, dim=1)
    dist = 2 - 2 * torch.mm(x, y.t())
    return dist
```

✗ low educational value

```
class Default(object):
    def __init__(self, vim):
        self._vim = vim
        self._bufnr = -1
        self._winid = -1
        self._winrestcmd = ''
        self._initialized = False
```

A good textbook is clear, self-contained, instructive, balanced

# The data recipe



# Step 1 · Filter the web




prompt: *“determine its educational value for a student whose goal is to learn basic coding concepts”*

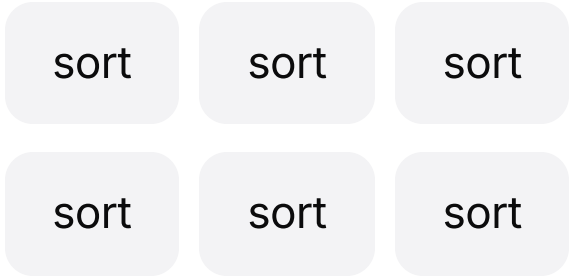
# Filtering alone helps

350M model · HumanEval pass@1

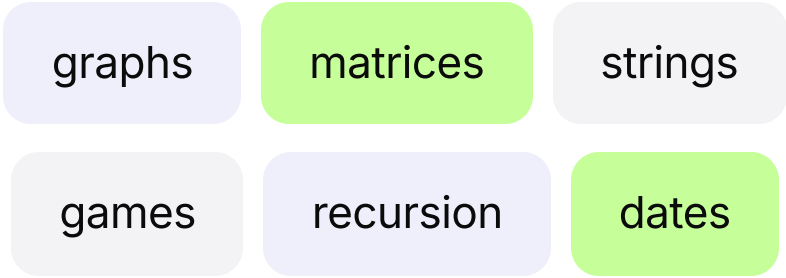


# Step 2 · Synthetic textbooks: diversity

GPT-3.5 · LLMs follow the most probable path




naive prompting




random topic + random audience

# A synthetic textbook page

“A matrix is said to be singular if its determinant is zero... Now, let's explore these concepts through examples.”

```
import numpy as np

def is_singular(A):
    det = np.linalg.det(A)
    return det == 0

A = np.array([[1, 2], [2, 4]])
print(is_singular(A))  # True
```

# Step 3 · CodeExercises

≈ 180M tokens · docstring → solution · diversity via function names

```
def valid_guessing_letters(word: str, guesses: List[str]) -> List[str]:
    """Returns a list of valid guessing letters, which are letters
    that have not been guessed yet and are present in the word."""
    valid_letters = []
    for letter in word:
        if letter not in guesses and letter not in valid_letters:
            valid_letters.append(letter)
    return valid_letters
```

# Model & training

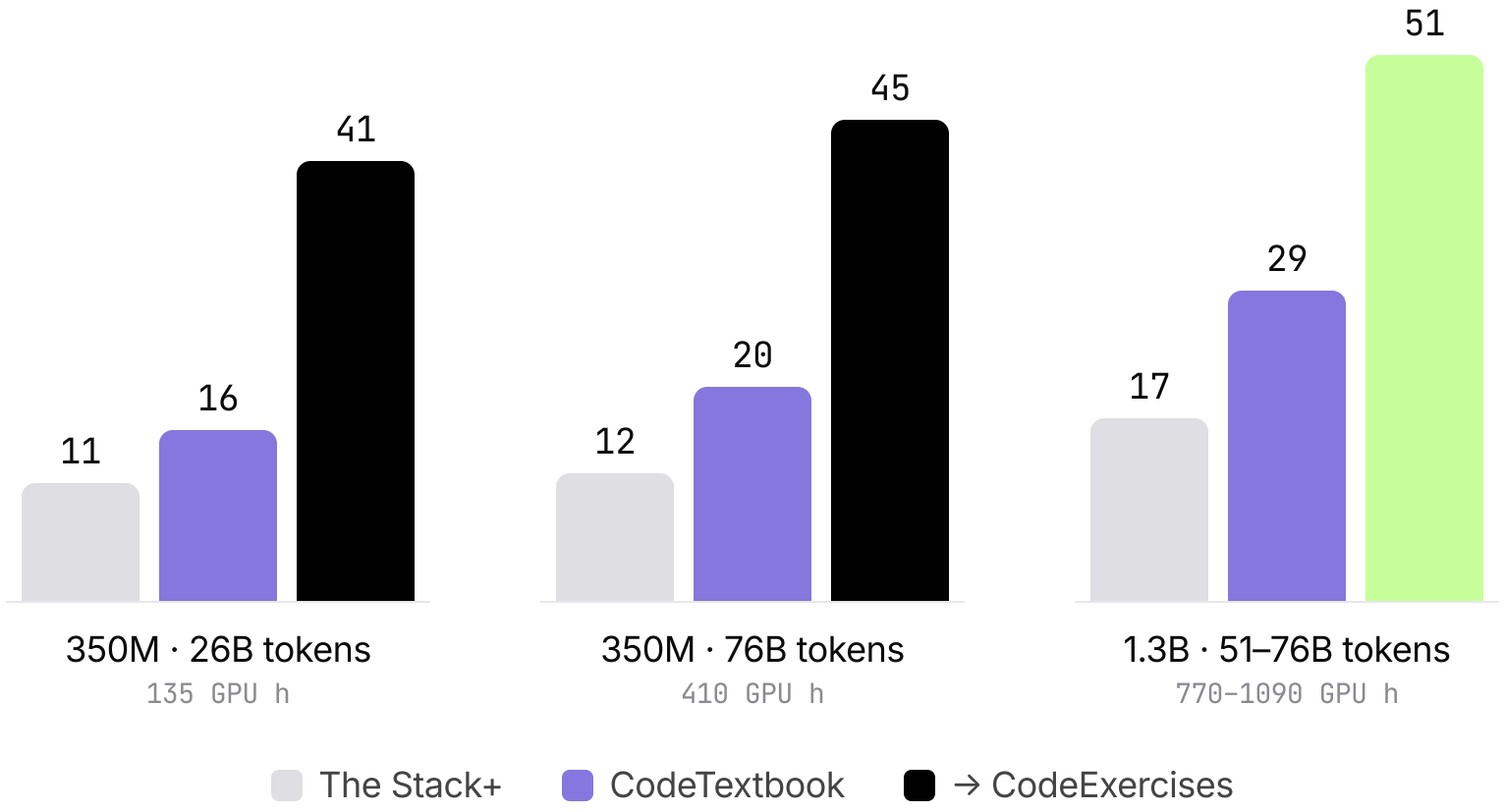
decoder-only transformer, deliberately plain

| | phi-1 | phi-1-small |
|---|---|---|
| Parameters | 1.3B | 350M |
| Layers | 24 | 20 |
| Hidden / MLP dim | 2048 / 8192 | 1024 / 4096 |
| Heads × dim | 32 × 64 | 16 × 64 |

FlashAttention | parallel MHA + MLP | RoPE | × FIM | × MQA

# The key experiment

HumanEval pass@1 (%) · Figure 2.1



180M tokens of exercises:

29% → 51%

# Emergent skills · understanding

finetuning on simple exercises unlocks skills not in the exercises

`phi-1`

✓ correct

follows every rule

`phi-1-base`

✕ rambles

endless `self.x = …`

`phi-1-small`

~ partial

gets the idea, wrong logic

prompt: Alice, Bob and Charles pick numbers... simulate 30 rounds

# Emergent skills · libraries

PyGame & Tkinter — absent from the exercises

| | knows the API | follows the logic |
|---|---|---|
| **phi-1** | ✓ | ✓ |
| phi-1-base | ✓ | ✗ |
| phi-1-small | ✗ | ✓ |

# Memorization? · 50 new problems

separate team · designed to be unusual · graded 0–10 by GPT-4

```
def sort_concat_square_deduplicate(list1, list2, my_threshold):
    """Sorts each list, concatenates them, squares the entries
    at even indices, filters out entries smaller than
    my_threshold and then removes duplicates."""
```

# Same ranking on new problems

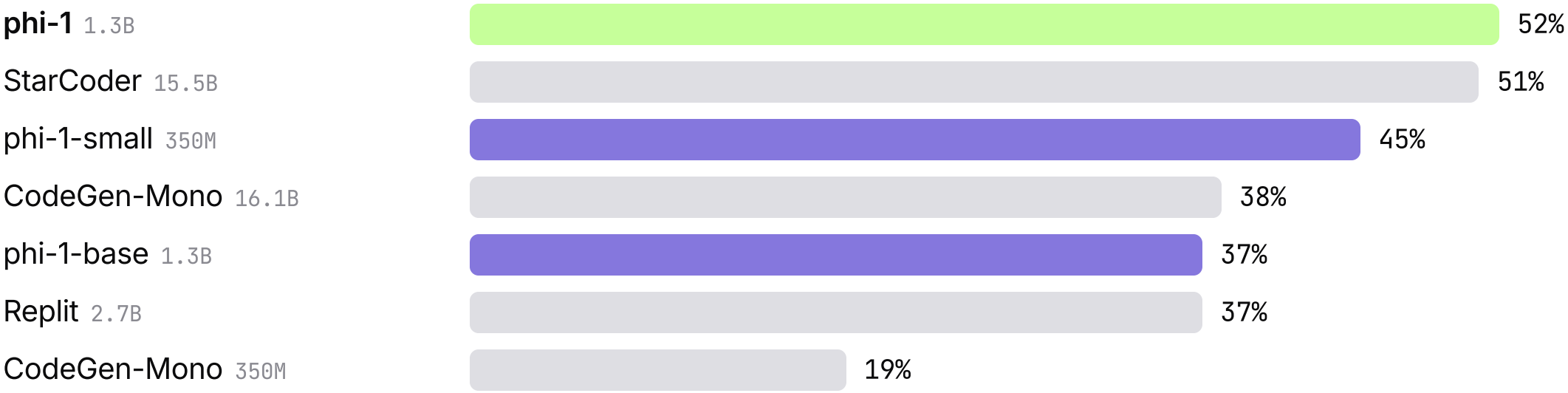


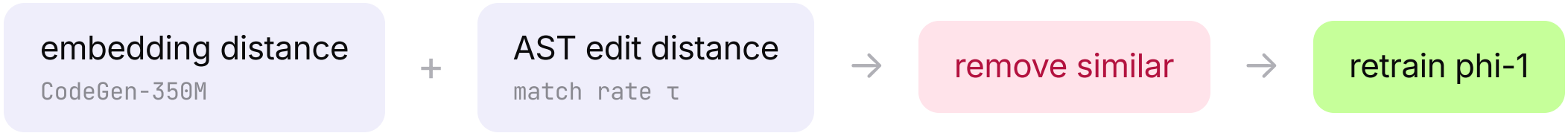

Table 2 · LLM-graded understanding score on 50 unconventional problems

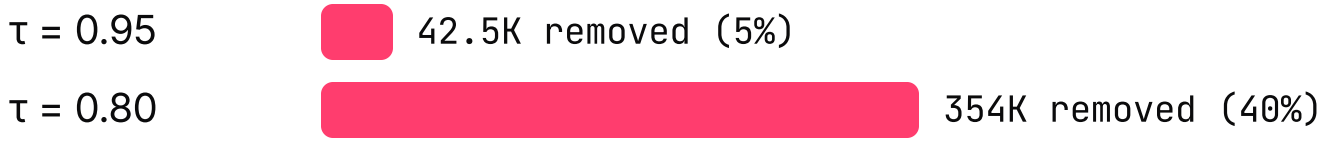
# n-gram overlap

| 13 | 4 | 0 |
| --- | --- | --- |
| -gram window | HumanEval hits | real matches |

all 4 were false positives — shared phrase “frequency of an integer…”

# Prune anything similar



# Still beats StarCoder

HumanEval pass@1 after retraining on pruned data

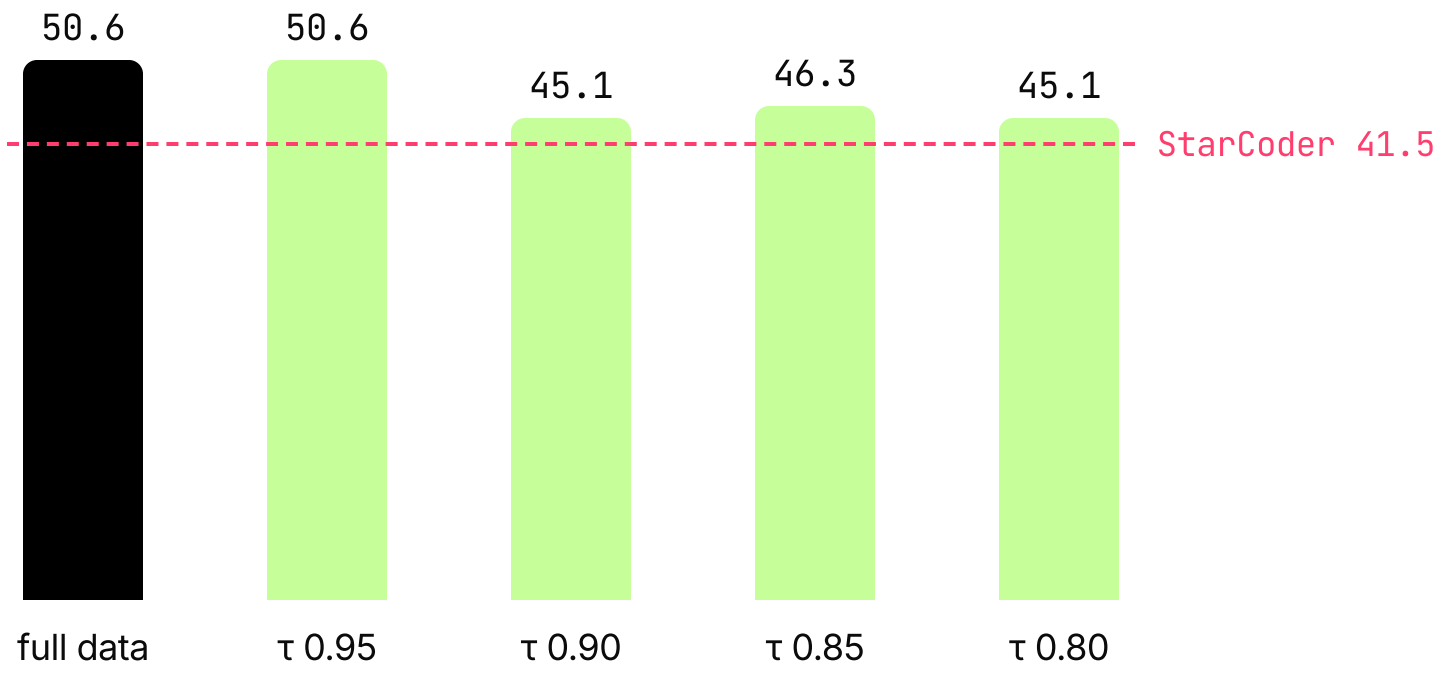



Table 3

# Limitations

scope

## Python only

narrow APIs, no complex apps

prompts

## long prompts fail

3 layers ✓ · 4 layers ✗ · + `import torch` ✓

language

## typos & wording

“x stays unchanged” → `x = 0`

reasoning

## counting & layout

extra text field, wrong buttons

# Open problems they name

01

## Coverage

all concepts, balanced

02

## Diversity

no repetition

03

## Measurement

how to quantify diversity?

04

## Ethics

LLMs curating data for LLMs

paper in one line: 10× smaller model · 100× less data · near-SOTA on code

Beyond the paper

# What happened next

the Phi family and its impact · 2023 → 2026

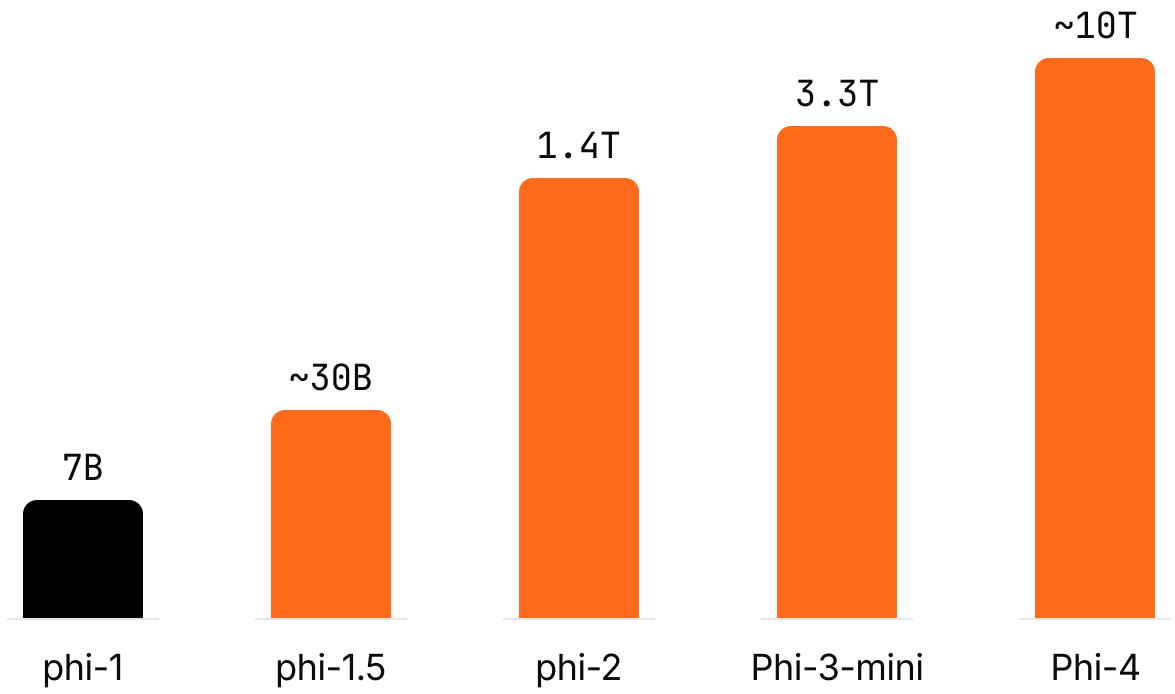
# The Phi family

| Jun 2023 | Sep 2023 | Dec 2023 | Apr 2024 | Dec 2024 | Apr 2025 | Mar 2026 |
|---|---|---|---|---|---|---|
| phi-1 | phi-1.5 | phi-2 | Phi-3 | Phi-4 | Phi-4-reasoning | Phi-4-r-vision |
| 1.3B · code this paper | 1.3B · common sense | 2.7B · general | 3.8B–14B on a phone | 14B · mostly synthetic | 14B · thinks | 15B · knows when to think |

Microsoft tech reports, Azure blog, VentureBeat

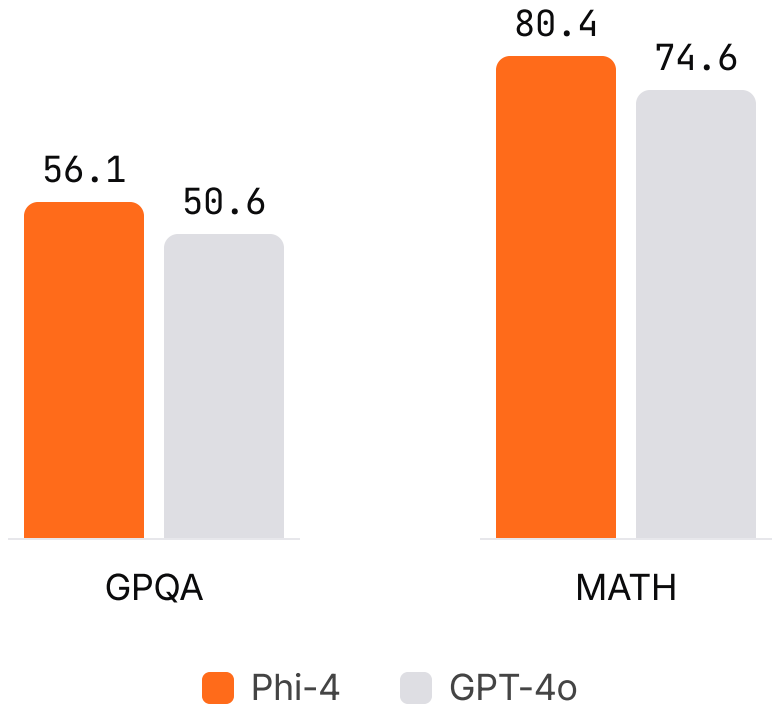
# Small model ≠ small data anymore

unique training tokens · log scale



# Phi-4 · student beats teacher

14B · ~400B synthetic tokens · vs its data generator GPT-4o




Phi-4 Technical Report, Dec 2024

# Phi today · October 2026

models

## No Phi-5 yet

Phi-4 line keeps specializing

where

## Windows, Azure, HF

Phi Silica on-device · MIT weights

people

## Bubeck → OpenAI

Oct 2024 · most co-authors stayed

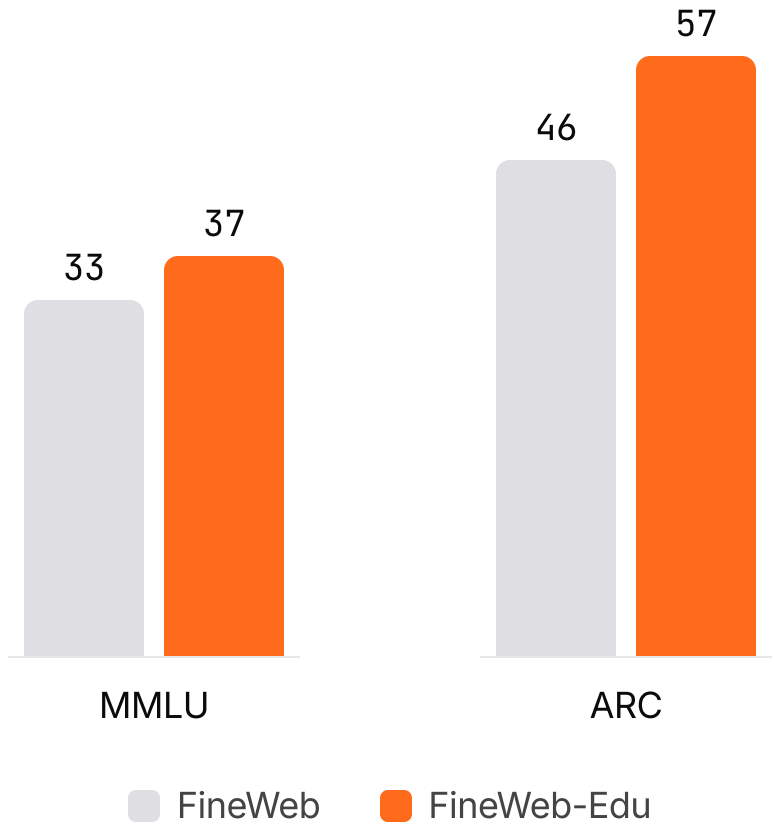
Microsoft

## Also MAI models

in-house large models next to Phi

# Impact · “educational value” filters

FineWeb-Edu (Hugging Face, 2024) · Llama-3-70B labels → classifier → keep score ≥ 3/5




Penedo et al., 2024 · arXiv 2406.17557

# Impact · synthetic data goes mainstream

open textbooks

## Cosmopedia → SmolLM

~25B tokens by Mixtral · data + code released

rewrite the web

## rephrase as QA, tutorials

WRAP · Nemotron-CC · BeyondWeb

distillation

## teacher writes the textbook

GPT-4o → Phi-4 · R1 → small reasoners

Hugging Face blog 2024 · NVIDIA Nemotron-CC · DatologyAI

# Critiques

Sep 2023 · parody

## “Pretraining on the Test Set Is All You Need”

1M-param phi-CTNL “beats everything”

community

## benchmarks » vibes?

strong scores, weaker in messy real use

Nature 2024

## model collapse

recursive training on own outputs degrades models

Schaeffer, arXiv 2309.08632 · Shumailov et al., Nature 2024

# Did the paper's claims hold up?

| claim in the paper | today |
|---|---|
| data quality bends scaling laws | ✓ widely accepted |
| LLM-generated training data works | ✓ standard practice |
| LLM-labelled quality filters | ✓ FineWeb-Edu, Llama 3 |
| tiny data is enough | ✗ Phi now uses trillions |
| benchmarks reflect real skill | ~ still debated |

# Takeaways

01

Data quality is a scaling axis

02

A small finetune can unlock skills

03

Check contamination seriously

# Thank you

Questions?

# Sources


**Gunasekar et al., 2023**
Textbooks Are All You Need · arXiv:2306.11644

**Kaplan et al., 2020**
Scaling Laws for Neural LMs · arXiv:2001.08361

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Phi-3 Technical Report · arXiv:2404.14219

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Phi-4-reasoning · arXiv:2504.21318

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Cosmopedia blog · huggingface.co/blog/cosmopedia

WEB **Shumailov et al., 2024**
AI models collapse… · Nature 631

**Eldan & Li, 2023**
TinyStories · arXiv:2305.07759

**Chen et al., 2021**
Evaluating LLMs Trained on Code · arXiv:2107.03374

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Phi-4 Technical Report · arXiv:2412.08905

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The FineWeb Datasets · arXiv:2406.17557

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Pretraining on the Test Set Is All You Need · arXiv:2309.08632

WEB **VentureBeat, Mar 2026**
Phi-4-reasoning-vision-15B